Duality in monoidal categories

based on joint work with T. Zorman

26.01.2026

Sebastian Halbig
Sebastian.Halbig@uni-marburg.de

University of Marburg

Advertisement: Upcoming seminar on TQC

New *online* seminar on *recent developments in topological quantum computing* (this combines in particular *tensor categories* and low-dimensional topology).

To get a better glimpse on this topic: Wait for Sung's talk!
Have a look at our website and mailing list:

https://sites.google.com/view/tqc-seminar

Our contact us: Nezhla Aghaei, Shawn Cui, Sebastian Halbig.

Two types of duality

Fix $(\mathcal{C}, \otimes, 1)$ a strict monoidal category.

Two types of duality

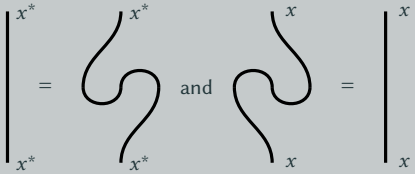
Fix $(\mathcal{C}, \otimes, 1)$ a strict monoidal category.

Rigid monoidal categories

$\mathcal{C}$ is *(right) rigid* if every $x \in \mathcal{C}$ admits a *right dual* $x^* \in \mathcal{C}$ together with two maps

$$\mathrm{ev}_x \colon\ x \otimes x^* \longrightarrow 1, \qquad \mathrm{coev}_x \colon\ 1 \longrightarrow x^* \otimes x$$

that satisfy the snake identities:



Two types of duality

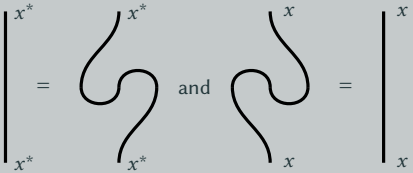
Fix $(\mathcal{C}, \otimes, 1)$ a strict monoidal category.

Rigid monoidal categories

$\mathcal{C}$ is *(right) rigid* if every $x \in \mathcal{C}$ admits a *right dual* $x^* \in \mathcal{C}$ together with two maps

$$\mathrm{ev}_x \colon\ x \otimes x^* \longrightarrow 1, \qquad \mathrm{coev}_x \colon\ 1 \longrightarrow x^* \otimes x$$

that satisfy the snake identities:



Fun application: If $\mathcal{C} = R\text{-}\mathsf{Mod}_{\mathrm{fin}}$ for a commutative ring R, then $\mathcal{C}$ is rigid iff R is semisimple.

Two types of duality

r-categories

$\mathcal{C}$ is an r-category if for every $x \in \mathcal{C}$ there is a dual $Dx \in \mathcal{C}$ such that

$$\mathcal{C}(x \otimes y, 1) \cong \mathcal{C}(y, Dx) \qquad \text{natural in } y \in \mathcal{C}$$

and the induced *dualising functor* $D\colon \mathcal{C}^{\mathrm{op}} \longrightarrow \mathcal{C}$ is an equivalence of categories.

Two types of duality

r-categories

$\mathcal{C}$ is an r-category if for every $x \in \mathcal{C}$ there is a dual $Dx \in \mathcal{C}$ such that

$$\mathcal{C}(x \otimes y, 1) \cong \mathcal{C}(y, Dx) \qquad \text{natural in } y \in \mathcal{C}$$

and the induced *dualising functor* $D\colon\ \mathcal{C}^{\mathrm{op}} \longrightarrow \mathcal{C}$ is an equivalence of categories.

Fact: Every rigid category is an r-category with dualising functor $x \longmapsto x^*$.

Two types of duality

r-categories

$\mathcal{C}$ is an r-category if for every $x \in \mathcal{C}$ there is a dual $Dx \in \mathcal{C}$ such that

$$\mathcal{C}(x \otimes y, 1) \cong \mathcal{C}(y, Dx) \qquad \text{natural in } y \in \mathcal{C}$$

and the induced *dualising functor* $D\colon\ \mathcal{C}^{\mathrm{op}} \longrightarrow \mathcal{C}$ is an equivalence of categories.

Fact: Every rigid category is an r-category with dualising functor $x \longmapsto x^*$.

Fun application: If $\mathcal{C} = R\text{-}\mathsf{Mod}_{\mathrm{fin}}$ for a commutative ring R, then $\mathcal{C}$ is an r-category if and only if R is quasi-Frobenius.
In this case $\{\text{injectives}\} = \{\text{projectives}\}$.

The main question

Tensor-hom adjunction for rigid and r-categories

Fix $x \in \mathcal{C}$. Depending on whether $\mathcal{C}$ is rigid or an r-category, we have specific shapes for the *tensor-hom adjunction*:

- **rigid:** $\mathcal{C}(x \otimes y, z) \cong \mathcal{C}(y, x^* \otimes z)$,
- **r-category:** $\mathcal{C}(x \otimes y, z) \cong \mathcal{C}(D^{-1}z \otimes x \otimes y, 1) \cong \mathcal{C}(y, D(D^{-1}z \otimes x))$.

The main question

The main question

Does the shape of the tensor-hom adjunction tell us whether $\mathcal{C}$ is rigid?

The main question

The main question

Does the shape of the tensor-hom adjunction tell us whether $\mathcal{C}$ is rigid?

More precisely, is $\mathcal{C}$ rigid if and only if $x \otimes - \dashv x^\vee \otimes -$ for all $x \in \mathcal{C}$?

The main question

The main question

Does the shape of the tensor-hom adjunction tell us whether $\mathcal{C}$ is rigid?

More precisely, is $\mathcal{C}$ rigid if and only if $x \otimes - \dashv x^\vee \otimes -$ for all $x \in \mathcal{C}$?

Our goal

Show that in representation theory there are natural examples of *non-rigid* r-categories $\mathcal{C}$ satisfying $x \otimes - \dashv Dx \otimes -$ for all $x \in \mathcal{C}$.

A brief look at $\mathfrak{sl}_2$-crystals

There will be a detailed talk on $\mathfrak{sl}_2$-crystals by Moaaz. Here's a very brief overview.

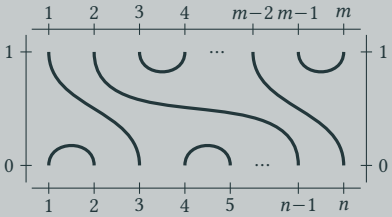
A brief look at $\mathfrak{sl}_2$-crystals

There will be a detailed talk on $\mathfrak{sl}_2$-crystals by Moaaz. Here's a very brief overview.

Construction of $\mathfrak{sl}_2$-crystals

1. Define CTL to be the category with objects $\mathbb{N}_0$ and morphisms

   $\mathrm{CTL}(n, m) = \mathrm{span}_{\mathbb{C}}\{\text{all planar string diagrams with } n \text{ inputs and } m \text{ outputs.}\}$



A brief look at $\mathfrak{sl}_2$-crystals

There will be a detailed talk on $\mathfrak{sl}_2$-crystals by Moaaz. Here's a very brief overview.

Construction of $\mathfrak{sl}_2$-crystals

1. Define CTL to be the category with objects $\mathbb{N}_0$ and morphisms

   $\mathrm{CTL}(n, m) = \mathrm{span}_{\mathbb{C}}\{\text{all planar string diagrams with } n \text{ inputs and } m \text{ outputs.}\}$

   Tensoring: pasting diagrams next to each other.
   Composition: vertical stacking plus the relations:

A brief look at $\mathfrak{sl}_2$-crystals

There will be a detailed talk on $\mathfrak{sl}_2$-crystals by Moaaz. Here's a very brief overview.

Construction of $\mathfrak{sl}_2$-crystals

1. Define CTL to be the category with objects $\mathbb{N}_0$ and morphisms

   $\mathrm{CTL}(n, m) = \mathrm{span}_{\mathbb{C}}\{\text{all planar string diagrams with } n \text{ inputs and } m \text{ outputs.}\}$

   Tensoring: pasting diagrams next to each other.
   Composition: vertical stacking plus the relations:

2. The category $\mathfrak{sl}_2(\mathbb{C})$-**crys** is the completion of CTL under taking direct sums and direct summands.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Semisimplicity

$\mathfrak{sl}_2(\mathbb{C})$-crys is semisimple and there is a bijection

$$\mathbb{N}_0 \longrightarrow \{\text{simples}\}/\sim, \qquad n \longmapsto [L_n],$$

such that $L_n \otimes L_m \cong \bigoplus_{k=0}^{\min\{n,m\}} L_{n+m-2k}$.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Lemma

We have a family of isomorphisms
$\phi_{x,y}\colon\ \mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes x, y) \longrightarrow \mathrm{Hom}_{\mathfrak{sl}}(x, L_1 \otimes y)$ *natural in* $x, y \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Lemma

We have a family of isomorphisms
$\phi_{x,y}\colon\ \mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes x, y) \longrightarrow \mathrm{Hom}_{\mathfrak{sl}}(x, L_1 \otimes y)$ *natural in* $x, y \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Note: The proof is quite insensitive to specifics of $\mathfrak{sl}_2(\mathbb{C})$-**crys**.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Lemma

We have a family of isomorphisms
$\phi_{x,y}\colon\ \mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes x, y) \longrightarrow \mathrm{Hom}_{\mathfrak{sl}}(x, L_1 \otimes y)$ *natural in* $x, y \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Proof(sketch).

1. **Semisimplicity:** We may assume $x = L_n$ and $y = L_m$ and $n \pm 1 = m$.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Lemma

We have a family of isomorphisms
$\phi_{x,y}\colon\ \mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes x, y) \longrightarrow \mathrm{Hom}_{\mathfrak{sl}}(x, L_1 \otimes y)$ *natural in* $x, y \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Proof(sketch).

1. **Semisimplicity:** We may assume $x = L_n$ and $y = L_m$ and $n \pm 1 = m$.
2. **Clebsch–Gordon:** The spaces $\mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes L_n, L_m)$ and $\mathrm{Hom}_{\mathfrak{sl}}(L_n, L_1 \otimes L_m)$ are spanned by projections $\pi^{\pm}$ and inclusions $\iota^{\mp}$.
   Signs: whether we consider the higher or lower summand.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Lemma

We have a family of isomorphisms
$\phi_{x,y}\colon\ \mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes x, y) \longrightarrow \mathrm{Hom}_{\mathfrak{sl}}(x, L_1 \otimes y)$ *natural in* $x, y \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Proof(sketch).

1. **Semisimplicity:** We may assume $x = L_n$ and $y = L_m$ and $n \pm 1 = m$.
2. **Clebsch–Gordon:** The spaces $\mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes L_n, L_m)$ and $\mathrm{Hom}_{\mathfrak{sl}}(L_n, L_1 \otimes L_m)$ are spanned by projections $\pi^{\pm}$ and inclusions $\iota^{\mp}$.
   Signs: whether we consider the higher or lower summand.
3. **Definition of ϕ:** The assignment $\pi^{\pm} \longmapsto \iota^{\mp}$ gives rise to an isomorphism
   $\phi_{n,m}\colon\ \mathrm{Hom}_{\mathfrak{sl}}(L_1 \otimes L_n, L_m) \longrightarrow \mathrm{Hom}_{\mathfrak{sl}}(L_n, L_1 \otimes L_m)$.
   Naturality comes from the commutativity of multiplication in $\mathbb{C}$. $\square$

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Theorem

The category $\mathfrak{sl}_2(\mathbb{C})$-**crys** *is a non-rigid* r*-category and* $x \otimes - \dashv x \otimes -$ *for all* $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Theorem

The category $\mathfrak{sl}_2(\mathbb{C})$-**crys** *is a non-rigid* r*-category and* $x \otimes - \dashv x \otimes -$ *for all* $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Note: Again, the proof does not rely on too specific statements about $\mathfrak{sl}_2(\mathbb{C})$-**crys**.

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Theorem

The category $\mathfrak{sl}_2(\mathbb{C})$-**crys** *is a non-rigid* r*-category and* $x \otimes - \dashv x \otimes -$ *for all* $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Proof(sketch).

- **Shape of the tensor hom adjunction:** Every $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys** is a direct summand of a direct sum of $L_1^{\otimes n}$. Thus, $x \otimes - \dashv x \otimes -$ follows from the previous lemma.

$\square$

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Theorem

The category $\mathfrak{sl}_2(\mathbb{C})$-**crys** *is a non-rigid* r*-category and* $x \otimes - \dashv x \otimes -$ *for all* $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Proof(sketch).

- **Shape of the tensor hom adjunction:** Every $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys** is a direct summand of a direct sum of $L_1^{\otimes n}$. Thus, $x \otimes - \dashv x \otimes -$ follows from the previous lemma.
- **Non-rigidity:** L_1 is self-dual and the defining relations of CTL imply that $\mathfrak{sl}_2(\mathbb{C})$-**crys** cannot be rigid.

$\square$

The duality-type of $\mathfrak{sl}_2(\mathbb{C})$-crys

Theorem

The category $\mathfrak{sl}_2(\mathbb{C})$-**crys** *is a non-rigid* r*-category and* $x \otimes - \dashv x \otimes -$ *for all* $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys**.

Proof(sketch).

- **Shape of the tensor hom adjunction:** Every $x \in \mathfrak{sl}_2(\mathbb{C})$-**crys** is a direct summand of a direct sum of $L_1^{\otimes n}$. Thus, $x \otimes - \dashv x \otimes -$ follows from the previous lemma.
- **Non-rigidity:** L_1 is self-dual and the defining relations of CTL imply that $\mathfrak{sl}_2(\mathbb{C})$-**crys** cannot be rigid.
- r**-category:** The dualising functor is on objects the identity. On endomorphisms of simples it must be bijective ($\rightsquigarrow$ fully-faithful).

$\square$